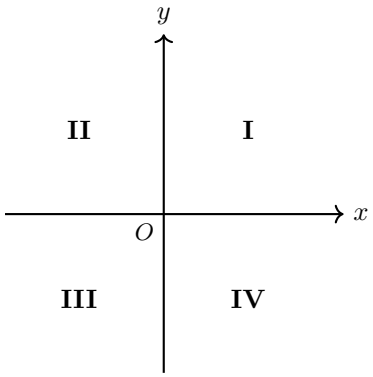
Lesson 1: The Cartesian Plane and Plotting Points

Reviewing axes, the origin, the four quadrants, and how ordered pairs locate points.

1. The Cartesian Plane

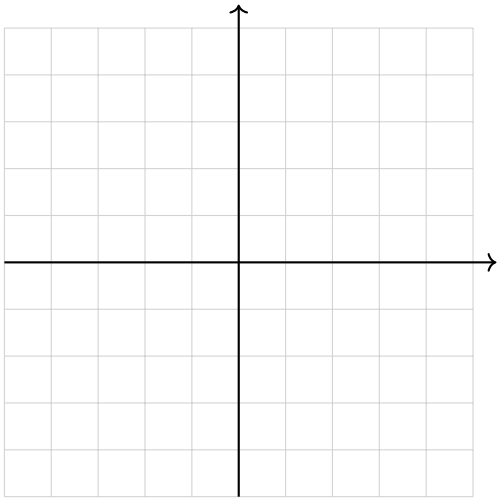
A **Cartesian coordinate system** is built from two perpendicular number lines called **axes**: the horizontal ***x*-axis** and the vertical ***y*-axis**. Together they form the **Cartesian plane**.
The axes cross at the **origin**, which is always located at $(0, 0)$.
The axes divide the plane into four regions called **quadrants**, numbered **I, II, III, IV** starting in the top-right region and moving **counter-clockwise**.



Example 1

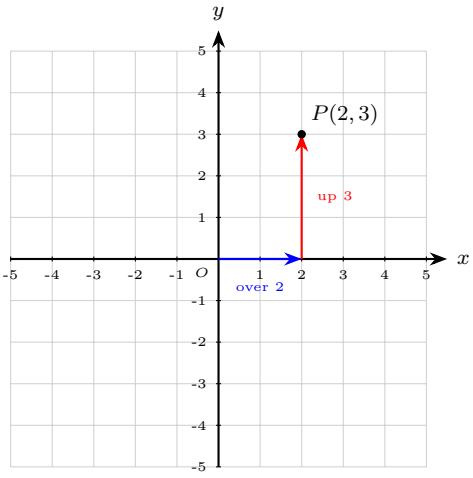
Complete the vocabulary, then label the quadrants.
(a) Fill in each blank.
The horizontal number line is called the _____-axis. The vertical number line is called the _____-axis. The point where the two axes cross is called the _____. This point has coordinates (___, ___).

(b) Write "I", "II", "III", or "IV" in each of the four regions.



2. Ordered Pairs

Every point on the plane is located by an **ordered pair** (x, y). The **first** number is the ***x*-coordinate**: how far to move **left or right** from the origin (positive is right, negative is left).
The **second** number is the ***y*-coordinate**: how far to move **up or down** (positive is up, negative is down). "Ordered" matters — swapping the two numbers gives a different point. Move **over** first, then **up or down**.
Worked demonstration: to plot $P(2, 3)$, start at the origin, move 2 units right, then 3 units up.

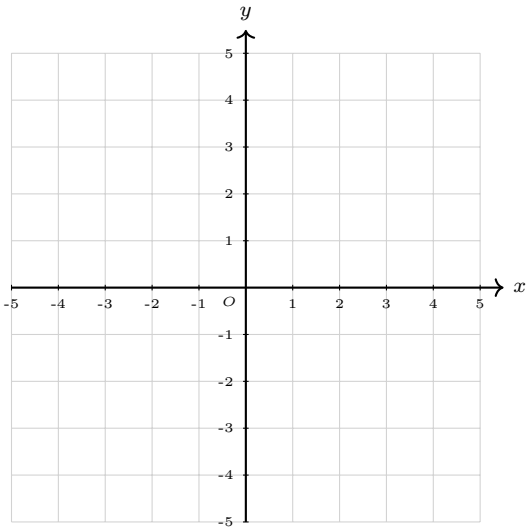


3. Plotting Points and Reading Their Coordinates

Reading coordinates is the reverse of plotting: the distance from the y-axis gives the x-coordinate, and the distance from the x-axis gives the y-coordinate. Example 2 below practises plotting; Example 3 reverses it.

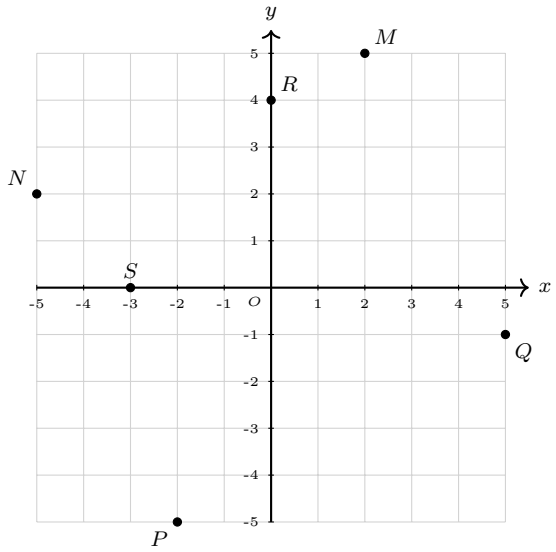
Example 2

Plot and label $A(2,4)$, $B(-3,1)$, $C(-1,-4)$, $D(4,-2)$ on the grid.



Example 3

State the coordinates of each labelled point.



M(_____) N(_____) P(_____)
Q(_____) R(_____) S(_____)

4. Which Quadrant Does a Point Lie In?

The **signs** of the coordinates give the quadrant without any plotting: I is $(+,+)$, II is $(-,+)$, III is $(-,-)$, IV is $(+,-)$. If $x=0$ the point lies on the y-axis; if $y=0$ it lies on the x-axis; $(0,0)$ is the origin and lies in no quadrant.

Example 4

Without plotting, state which quadrant each point lies in. If it lies on an axis, say which one.

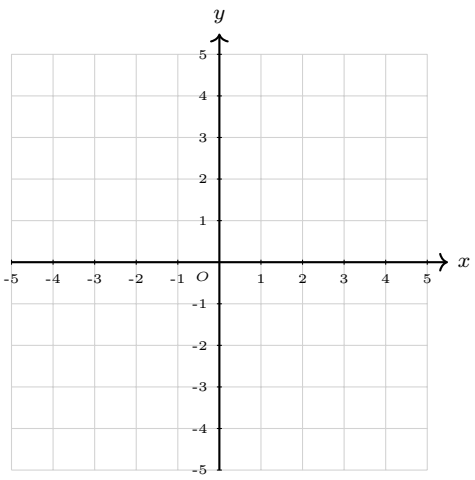
(a) $(5,3)$ (b) $(-4,2)$ (c) $(-3,-5)$ (d) $(6,-1)$ (e) $(0,-7)$ (f) $(-8,0)$

5. Plotting Several Points and Recognizing a Pattern

Several related points plotted on one grid often reveal a pattern, such as lining up in a straight row — the first step toward a **relation**, the topic of the next lesson. Two quick patterns to watch for: points with the **same y-coordinate** lie in a **horizontal row**, and points with the **same x-coordinate** lie in a **vertical column**.

Example 5

Consider the points $(-4,-4)$, $(-2,-2)$, $(0,0)$, $(2,2)$, and $(4,4)$. **(a)** Plot all five points on the grid.



(b) What do you notice about the way these five points are arranged?

(c) Other than the origin, which two quadrants contain these points?

Lesson 2: Growing Patterns and Types of Variables

Describing a growing pattern in words, in a table of values, and as points on a grid.

1. Growing Patterns

A **growing pattern** is a sequence of figures or numbers in which each step grows by the *same fixed amount* every time; we say the pattern grows by a **constant amount**. In the tile pattern below, look for what each new figure adds.

Example 1

The pattern below is built from small square tiles arranged in two rows.

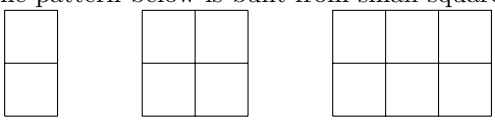


Figure 1 Figure 2 Figure 3

(a) Draw Figure 4 and Figure 5 of the pattern in the space below.

(b) How many tiles would be in Figure 6? (Continue the pattern.)

(c) Describe in words how this pattern grows from one figure to the next.

2. Representing a Pattern in a Table of Values

A **table of values** stores the figure number in one row and the resulting value in the other, which makes the pattern easy to read, extend, and later plot. For the tile pattern above: $n = 1, 2, 3$ gives $T = 2, 4, 6$.

Example 2

Continue working with the tile pattern from Example 1.

(a) Complete the table of values below for Figures 1 through 6.

Figure Number, n	1	2	3	4	5	6
Number of Tiles, T	2					

(b) How many tiles are added each time you move to the next figure?

(c) Continue the pattern to find the number of tiles in Figure 10.

3. Independent and Dependent Variables

If notebooks cost \$2 each, the number you buy is the **independent variable** (you choose it) and the total cost is the **dependent variable** (it results from that choice). The independent variable goes in the first row of a table and on the horizontal (x) axis; the dependent variable goes on the vertical (y) axis. For the tiles, n is independent and T is dependent.

Example 3

For each situation, identify the independent variable and the dependent variable.

(a) The temperature of an oven while it preheats, based on the number of minutes since it was turned on.

Independent: _____
Dependent: _____

(b) The distance a cyclist has travelled, based on the number of hours cycling at a steady speed.

Independent: _____
Dependent: _____

(c) The height remaining on a candle, based on the number of hours it has been burning.

Independent: _____
Dependent: _____

(d) The total cost of a taxi ride, based on the number of kilometres travelled.

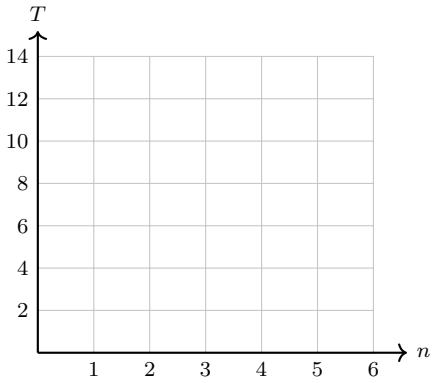
Independent: _____
Dependent: _____

4. Plotting a Pattern's Table of Values

Each column of a table of values is an ordered pair in disguise: the independent value becomes the x-coordinate and the dependent value becomes the y-coordinate. Because these values are positive whole numbers, only Quadrant I is needed. When a pattern grows by a **constant amount**, its points line up in a straight line — we call it a **linear** pattern.

Example 4

Use your completed table of values for the tile pattern from Example 2.



(a) Plot all six ordered pairs (n, T) on the grid shown.

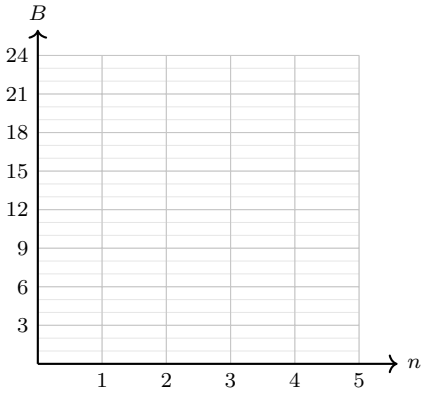
(b) Do the plotted points appear to line up in a straight pattern?

Example 5

A garden club builds a row of raised flower beds. The first bed uses 5 boards, and each additional bed needs 4 more boards.
(a) Complete the table of values below for beds 1 through 5.

Bed Number, n	1	2	3	4	5
Number of Boards, B	5				

(b) Describe in words how the number of boards grows from one bed to the next.



(c) Plot the ordered pairs (n, B) from your table on the grid shown.

(d) Do the plotted points appear to line up?

Lesson 3: Describing Relationships Between Two Quantities

Relations as ordered pairs, and the tables, mapping diagrams, and graphs that picture them.

1. What Is a Relation?

A **relation** is a set of ordered pairs connecting the inputs of one quantity to the outputs of another: hours slept and alertness, pizza size and price, pages read and time spent. Each ordered pair (x, y) matches one input x with its output y, so the list of ordered pairs *is* the relation in its most basic form.

Example 1

A vending machine sells juice boxes for \$2 each. Let n be the number bought and C the total cost, in dollars.
(a) Determine the total cost for $n = 1, 2, 3$, and 4 juice boxes.

(b) Write this relation as a set of ordered pairs (n, C).

2. Representing a Relation with a Table of Values

A **table of values** is an organized way of writing the ordered pairs: inputs in one row, each matching output directly below it (or beside it, when the table stands up). Reading straight down a column gives you one ordered pair.

Example 2

Each tricycle in a bike-shop display has 3 wheels. The table shows tricycles, t, and total wheels, W.

t	1	2	3	4
W	3	6	9	12

(a) Write this relation as a set of ordered pairs (t, W).

(b) State the independent variable and the dependent variable.

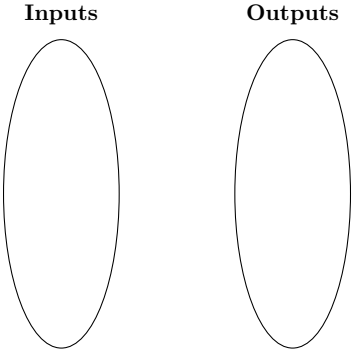
3. Representing a Relation with a Mapping Diagram

A **mapping diagram** shows a relation with two ovals joined by arrows: inputs in the left oval, outputs in the right oval. The arrows always point **from** the input **to** the output — the same "independent to dependent" direction you already know.

Example 3

Each chair at a banquet table has 4 legs. The table shows chairs, c, and total legs, L.

c	2	3	4	5
L	8	12	16	20



(a) Write this relation as a set of ordered pairs (c, L).

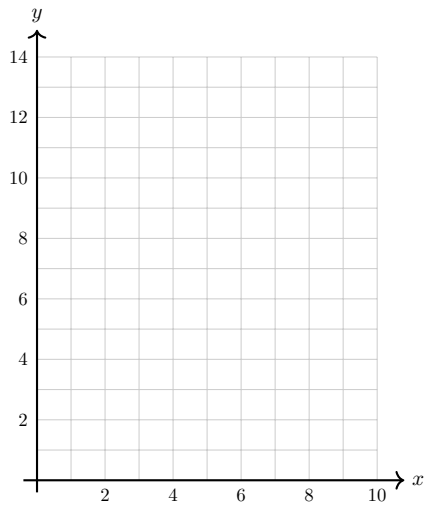
(b) Represent the relation with the mapping diagram at left: write the inputs in the left oval and the outputs in the right oval, then draw an arrow from each input to its matching output.

4. Representing a Relation with a Graph

A relation can also be pictured as a **graph**. Each ordered pair (x, y) becomes one point: the input is measured along the horizontal x-axis and the output along the vertical y-axis — independent horizontal, dependent vertical, exactly as before.

Example 4

Consider the relation given by the ordered pairs (x, y): $(1, 5),\ (2, 7),\ (3, 9),\ (4, 11)$.



(a) Complete the table of values below.

x	1	2	3	4
y				

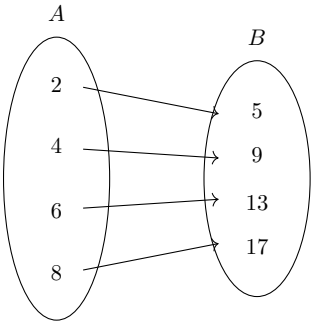
(b) Plot the four ordered pairs on the grid at left.

5. Same Relation, Three Pictures

A table of values, a mapping diagram, and a graph can all describe the **exact same relation** — they are three pictures of one set of ordered pairs. Find the ordered pairs and you can build any one representation from any other.

Example 5

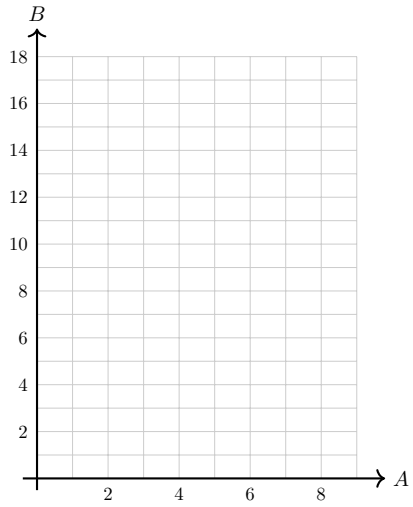
The mapping diagram below shows a relation between two sets of numbers, A and B.



(a) Write this relation as a set of ordered pairs (A, B).

(b) Complete a table of values for this relation.

A				
B				



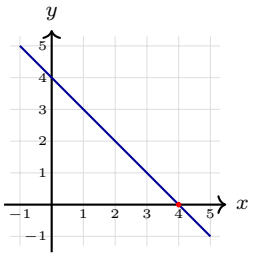
(c) Plot the ordered pairs from part (a) on the grid at left to represent this relation as a graph.

(d) In one sentence, explain why the mapping diagram, the table, and the graph all describe the same relation.

Lesson 4: Where Graphs Meet the Axes: x- and y-Intercepts

Reading the points where a graph crosses each axis, from graphs, tables, and real situations.

1. What an Intercept Is

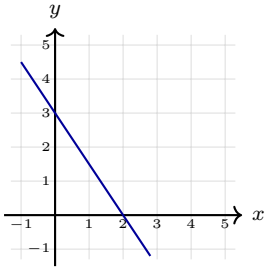


An **x-intercept** is a point where a graph crosses the x-axis; a **y-intercept** is a point where it crosses the y-axis. Every point on the x-axis has $y = 0$, so an x-intercept looks like $(\,\square,\, 0\,)$; every point on the y-axis has $x = 0$, so a y-intercept looks like $(\,0,\, \square\,)$. So **an x-intercept is the point where $y = 0$, and a y-intercept is the point where $x = 0$.** On a graph, read off the point where the curve meets the axis — the line at left meets the x-axis at $(4, 0)$.

Examples 1 and 2

Each graph below shows a line.

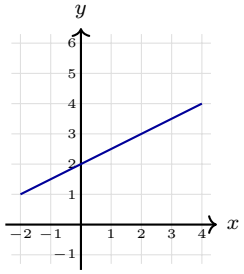
Example 1



(a) State the coordinates of the point where the graph crosses the x-axis.

(b) What is the value of y at that point?

Example 2



(a) State the coordinates of the point where the graph crosses the y-axis.

(b) What is the value of x at that point?

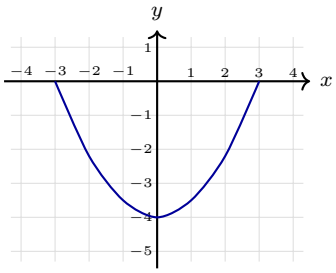
2. Several Intercepts, and Intercepts from a Table

A graph can meet the x-axis more than once, so list every spot where the graph touches an axis. A table of values works the same way without a picture: the column (or row) where $x = 0$ gives the y-intercept, and the column (or row) where $y = 0$ gives the x-intercept.

Examples 3 and 4

Example 3 shows a curve; Example 4 shows several points on a relation.

Example 3



(a) State the coordinates of both x-intercepts.

(b) State the coordinates of the y-intercept.

Example 4

x	-3	-2	-1	0	1
y	-1	0	1	2	3

(a) Which ordered pair is the y-intercept?

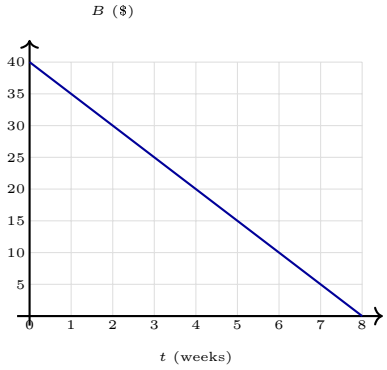
(b) Which ordered pair is the x-intercept?

3. Intercepts in Real Situations

When a graph models a real situation, the y-intercept usually gives the **starting value** (before any time has passed) and the x-intercept usually tells **when a quantity reaches zero**.

Example 5

Malik loads \$40 onto a gift card and spends it steadily; B is the balance after t weeks.



(a) What is the y-intercept of the graph? What does it represent here?

(b) What is the x-intercept of the graph? What does it represent here?

(c) Use the graph to estimate the balance remaining after 4 weeks.

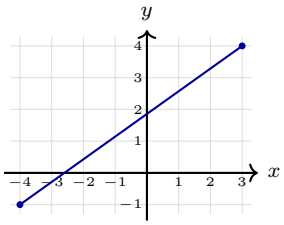
(d) Use the graph to estimate after how many weeks the balance is \$10.

Lesson 5: Domain and Range of a Relation, Part 1: Reading From a Graph

Reading how far a graph stretches left to right and bottom to top, discrete or continuous.

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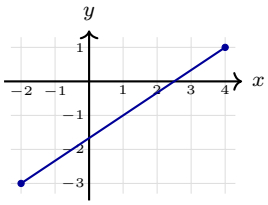
1. Domain: How Far the Graph Reaches Left to Right



The **domain** of a relation is the set of all possible x-values — the *inputs* the relation uses. To read it from a graph, find the graph's **leftmost** point and its **rightmost** point; the domain is every x-value from one to the other. The graph at left starts at $x = -4$ and ends at $x = 3$, and every x-value in between is used somewhere on the line, so its domain is: *x can be any value from -4 to 3.*

Example 1

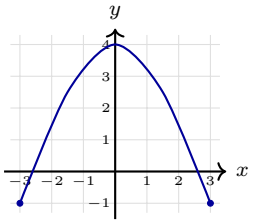
The graph below shows a line segment.



(a) State the domain of the graph.

(b) Could $x = 5$ be part of this relation? Explain.

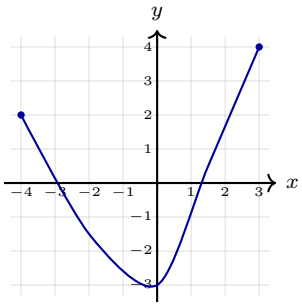
2. Range: How Far the Graph Reaches Bottom to Top



The **range** of a relation is the set of all possible y-values — the *outputs* the relation produces. To read it from a graph, find the graph's **lowest** point and its **highest** point; the range is every y-value from one to the other. The lowest point at left is $y = -1$ and the highest is the peak at $y = 4$, so the range is: *y can be any value from -1 to 4.*

Example 2

The graph below shows a curve.



(a) State the range of the graph.

(b) Could $y = 5$ be part of this relation? Explain.

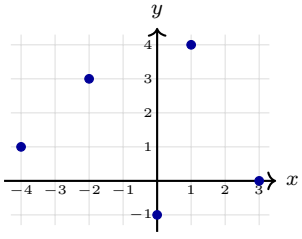
3. Discrete Graphs vs. Continuous Graphs

A **discrete** graph is a set of separate dots that are not joined; only the plotted points belong to the relation, so its domain and range are written as a **list** of the exact values used ("x can be -3, 0, or 4"). A **continuous** graph is an unbroken line or curve with no gaps, so every point along it counts and we describe its domain and range as everything **from** the lowest value **to** the highest value.

Example 3

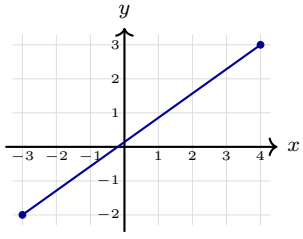
For each graph, state whether it is discrete or continuous, then state its domain and range.

(a)



Type: _____
D: _____
R: _____

(b)



Type: _____
D: _____
R: _____

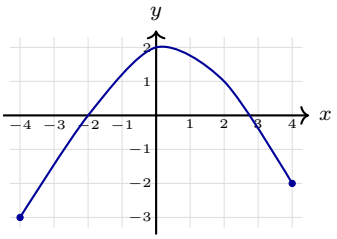
Reminder. A discrete answer is a list of the dots' values; a continuous answer runs "from ___ to ___." Count squares carefully before writing each endpoint.

4. Curves That Bend More Than Once

On a curve that bends more than once, the highest or lowest point is not always at one of the two endpoints. Always scan the **whole** graph, not just its ends, before stating the domain and range.

Example 4

The graph below shows a continuous curve.



(a) State the domain of the graph.

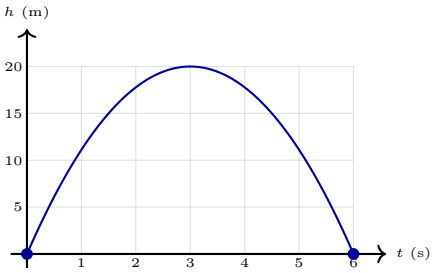
(b) State the range of the graph. (Hint: look at the whole curve, not just the two ends.)

5. Domain and Range in Everyday Graphs

When a graph models a real situation, the domain and range are limited to values that make sense: on a height-versus-time graph, **time cannot be negative**, because the situation only exists once it begins. Always check what is happening in real life before writing the domain and range.

Example 5

A toy rocket is launched straight up from the ground. The graph shows the rocket's height, h metres, after t seconds. It peaks at 20 m when $t = 3$ s and lands at $t = 6$ s.



(a) State the domain of the graph (in seconds).

(b) State the range of the graph (in metres).

(c) Explain why the domain of this graph could not include negative values of t.

Lesson 6: Domain and Range of a Relation, Part 2

Finding domain and range from ordered pairs and tables, then writing them in set notation.

1. Recall: Domain and Range from a Graph

The **domain** is the complete set of input (x) values a relation uses; the **range** is the complete set of output (y) values it produces. On a graph we found these by shining an imaginary light down (domain) or across (range). Today the relation arrives already written as ordered pairs or in a table, so we read the values directly instead.

2. Domain and Range from a Set of Ordered Pairs

Every ordered pair has the form (x, y): the **domain** is every x-coordinate that appears, the **range** is every y-coordinate that appears, and a repeated value is written only *once*. For $(2, 5)$, $(3, 5)$, $(4, 8)$, $(6, 1)$ the x's are $2, 3, 4, 6$ (no repeats) so the domain is $2, 3, 4, 6$; the y's are $5, 5, 8, 1$, and since 5 repeats the range is $1, 5, 8$. List the values from smallest to largest — it makes repeats easy to spot.

Example 1

List the domain and range of each set of ordered pairs.

(a) $(1, 4)$, $(2, 6)$, $(3, 4)$, $(5, 9)$

(b) $(7, 2)$, $(7, 5)$, $(7, 8)$

3. Domain and Range from a Table of Values

A table lists the same information as a set of ordered pairs, just organized into two rows instead of parentheses. The same rule applies: the **domain** is every value in the x-row and the **range** is every value in the y-row, each written once.

Example 2

State the domain and range of each relation given by a table of values.

(a)

x	0	1	2	3	4
y	10	8	6	4	2

(b)

x	-3	-1	1	3
y	5	2	5	2

4. Formal Set Notation for Domain and Range

Mathematicians write these lists inside curly braces { } to show they form a **set** — a collection of values with no repeats. The demonstration above becomes $D = \{2, 3, 4, 6\}$, $R = \{1, 5, 8\}$. Some relations have infinitely many values and cannot be listed, so we describe them with an **inequality** instead. An **arrow** on the end of a graph means it **keeps going forever**: a ray that starts at $(2, 1)$ and rises forever up and to the right has every y-value 1 or greater, so its range is $y \geq 1$ (read $\geq$ as "is greater than or equal to"; $\leq$ is "is less than or equal to"). A continuous piece with solid dots at both ends works the same way: "x from -3 to 4" (Lesson 5) is written $-3 \leq x \leq 4$. Recognize when a plain list works (a finite, discrete relation) and when an inequality is the better tool (a continuous or unending relation).

Example 3

Write the domain and range using set notation (or an inequality where a list is impossible).

(a) The ordered pairs $(2, 5),\ (4, 5),\ (6, 9),\ (8, 3)$.

(b) The table below.

x	-4	-2	0	2
y	7	7	7	7

(c) A ray starts at $(0, 0)$ and rises forever up and to the right, with an arrow at the end. Write the domain and the range using inequalities.

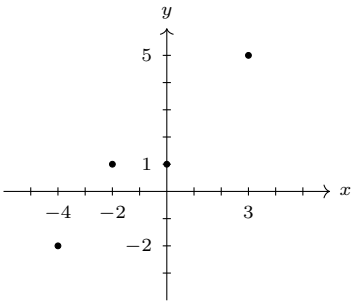
5. Combining Skills

We can now find the domain and range of a relation however it is presented — ordered pairs, a table, or a graph — and write the answer in proper set notation.

Example 4

State the domain and range, using set notation, of each relation below.

(a) $(-3, 8),\ (-1, 8),\ (1, 2),\ (3, -4)$



(b) The table below shows five ordered pairs.

x	1	2	3	4	5
y	3	6	9	12	15

(c) The graph at the left shows four points.

Example 5

Deshawn earns \$12 per hour at his part-time job. He can work at most 10 hours per week at this job. The table shows his earnings, E dollars, for various numbers of hours worked, h.

h (hours)	2	4	6	8	10
E (dollars)	24	48	72	96	120

(a) State the domain and range of this relation using set notation.

(b) Explain why h cannot be greater than 10 in this situation.

(c) Could Deshawn earn \$144 in one week at this job? Explain.

Lesson 7: Using Technology to Graph and Explore Relations

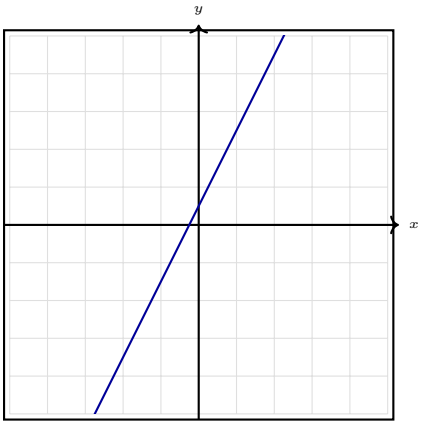
Graphing with a calculator or Desmos: setting a window, finding intercepts, and reading domain and range.

1. Entering a Relation and Setting a Window

The **viewing window** is a picture frame laid over the plane: only the part of the graph inside the frame appears on screen. Four numbers control it — **Xmin/Xmax** (smallest and largest x shown) and **Ymin/Ymax** (smallest and largest y shown). The **standard window** is $x : [-10, 10]$, $y : [-10, 10]$. *TI-84:* 1) press `Y=` and type the right side of the equation (solve for y first); 2) press `WINDOW` to edit Xmin, Xmax, Ymin, Ymax; 3) press `GRAPH`. *Desmos:* 1) type the equation on an empty line of the expression list; 2) the graph appears at once — drag or scroll to zoom; 3) click the wrench icon to type exact window values.

Example 1

Deshawn graphs $y = 2x + 1$ and leaves the standard window, $x : [-10, 10]$, $y : [-10, 10]$, in place.



(a) State the window shown: $x : [____, ____]$, $y : [____, ____]$.

(b) Does this window show the line clearly, with room to spare on every side? Explain.

(c) From the screen, estimate the x-intercept and the y-intercept.

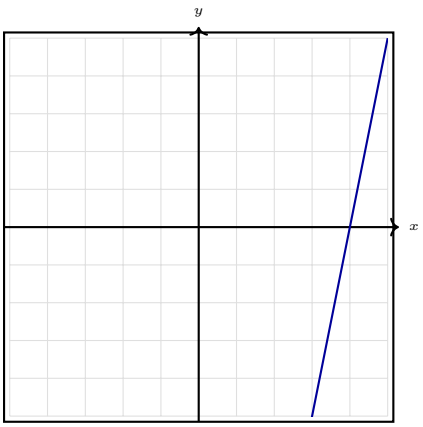
2. Adjusting the Window When Needed

If an intercept or an important part of the graph is missing, either **type a custom window** (`WINDOW` on a calculator, the wrench icon in Desmos) or **zoom out** (`ZOOM` then `3`; scroll or pinch in Desmos). Best strategy: estimate the intercepts first, then set Xmin/Xmax and Ymin/Ymax comfortably beyond them.

Example 2

Aisha graphs $y = 5x - 40$. Screen 1 uses the standard window; Screen 2 uses an adjusted window.

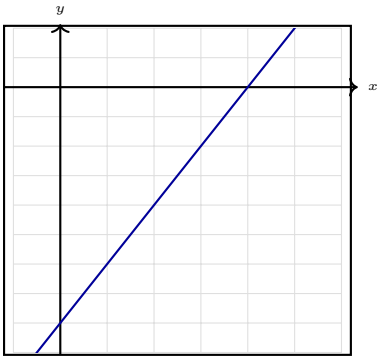
(a) What problem do you notice with Screen 1?



Screen 1: standard window

(b) Determine a better window that shows both intercepts comfortably. State Xmin, Xmax, Ymin, Ymax.

(c) Does Screen 2 show the full shape of the line, including both intercepts, more clearly than Screen 1? Explain.



Screen 2: adjusted window

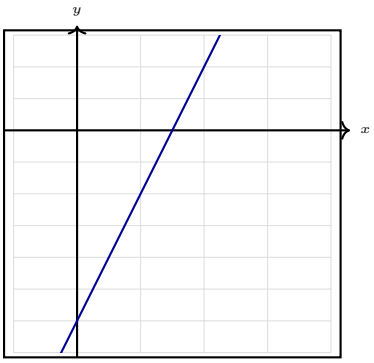
3. Using Technology to Find Intercepts

An x-intercept is where the graph crosses the x-axis ($y = 0$); a y-intercept is where it crosses the y-axis ($x = 0$). Built-in tools: the **zero** (root) feature locates an x-intercept; **trace** or **table** moves along the curve or lists (x, y) pairs, so $x = 0$ gives the y-intercept; the **value** feature evaluates y at any chosen x.

Example 3

The screen shows $y = 4x - 12$ using the window $x : [-2, 8]$, $y : [-14, 6]$.

(a) Use the zero (or trace) feature to determine the x-intercept.



(b) Use the trace, table, or value feature (at $x = 0$) to determine the y-intercept.

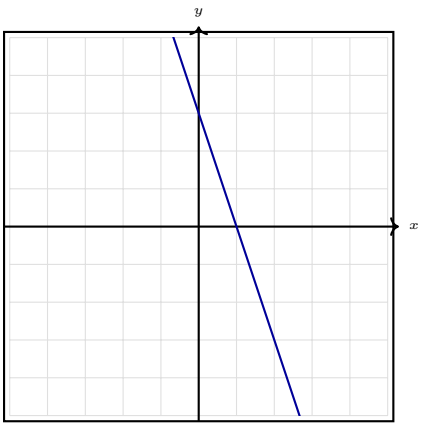
(c) Write both intercepts as coordinate pairs.

4. Reading Domain and Range Off the Screen

Domain is the set of x-values (scan left to right); range is the set of y-values (scan bottom to top). If the graph runs off the edges of the window it keeps going forever in that direction; if it visibly stops or turns around, the domain or range is limited. A straight, slanted line (not horizontal or vertical) runs off the screen at both ends, so its domain *and* range are both **all real numbers**.

Example 4

The screen shows $y = -3x + 6$ using the standard window, $x : [-10, 10]$, $y : [-10, 10]$.



(a) Does the line stop anywhere inside the screen, or does it run off the edges of the window in both directions?

(b) State the domain of this relation.

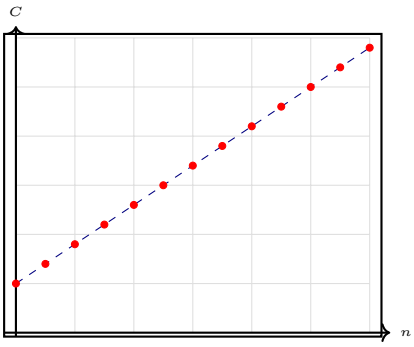
(c) State the range of this relation.

5. A Real-World Relation Explored with Technology

When a relation models a real situation, not every input is allowed — a number of items ordered cannot be negative. After technology graphs the relation and finds the intercepts, the *context* decides the domain and range: some situations allow only whole numbers (items ordered), others allow any value in between (hours).

Example 5

An online store charges a \$5 flat shipping fee plus \$2 per candle, so the total cost C of n candles is $C = 5 + 2n$. The window is $x : [0, 12]$, $y : [0, 30]$; red dots mark whole-number n.



(a) Use the trace, table, or value feature to determine the C-intercept (the cost when $n = 0$). What does it represent here?

(b) Algebraically determine the n-intercept. Does this value make sense as a number of candles? Explain.

(c) Use technology to determine the cost of ordering 6 candles.

(d) Use technology (or algebra) to determine how many candles were ordered if the total cost was \$23.

(e) State an appropriate domain and range for this relation, given the context.

Lesson 8: Introducing Functions: The Vertical Line Test

Every input gets exactly one output — and a fast visual check for graphs.

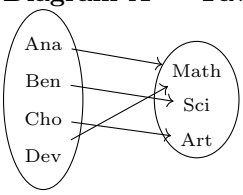
1. From Relations to Functions

A **relation** pairs the inputs of one quantity (the **domain**) with the outputs of another (the **range**); it may pair an input with any number of outputs. A **function** is the special relation in which every input is paired with exactly one output. If even one input is paired with two *different* outputs, the relation is **not** a function. In a **mapping diagram** just count arrows *leaving* each input: exactly one arrow from every input → function; two or more from any input → not a function. (Arrows pointing *into* an output never matter — an output may be reused as often as you like.)

Example 1

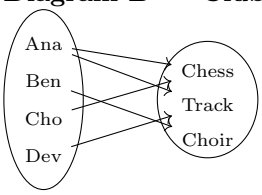
Four students were asked two questions. Diagram A gives each student's *one* favourite subject; Diagram B gives every club each student joined.

Diagram A — Favourite Subject



(a) Count the arrows leaving each student. Is this relation a function? Explain.

Diagram B — Clubs Joined



(b) Count the arrows leaving each student. Is this relation a function? Explain.

2. Checking a List of Ordered Pairs or a Table

The one-output rule does not change; only the way you look for it does. Scan the x-values: if the **same** x-value appears twice with two **different** y-values, it is **not** a function. A repeated y-value is never a problem, and a repeated pair listed twice is not either.

Example 2

Decide whether each relation is a function, and justify by checking the x-values.

(a) $(1,4)$, $(2,7)$, $(3,4)$, $(5,9)$

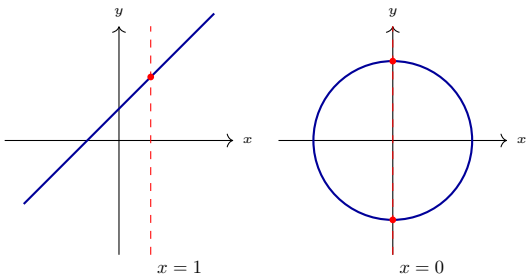
(b) $(2,6)$, $(3,8)$, $(2,11)$, $(4,8)$

(c)

x	-1	0	1	-1
y	5	5	5	5

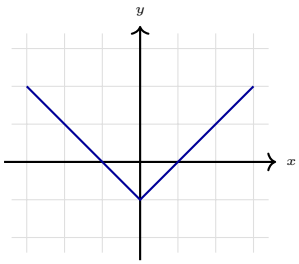
3. The Vertical Line Test

Vertical Line Test: if *any* vertical line crosses the graph **more than once**, the relation is **not** a function — that line has found one input with two different outputs. If *every* vertical line crosses **at most once**, the graph **is** a function. Below left, the line $x = 1$ meets the graph once (function). Below right, $x = 0$ meets the circle twice, at $y = 2.5$ and $y = -2.5$ (not a function).

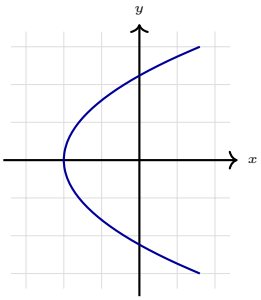


Example 3

Draw two or three vertical lines of your own on each grid, then decide.



(a) _____ Explain how you know.



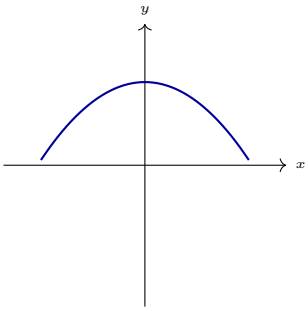
(b) _____ Explain how you know.

4. Classifying Several Graphs at Once

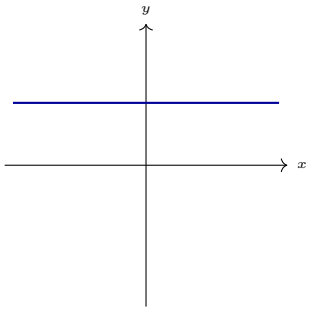
Run the test mentally from left to right across each graph. Closed curves (circles, ellipses, loops) and sideways parabolas always fail; straight lines that are not vertical, and ordinary parabolas opening up or down, always pass.

Example 4

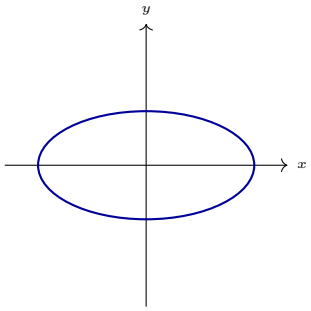
Use the vertical line test to classify each graph (write Yes or No).



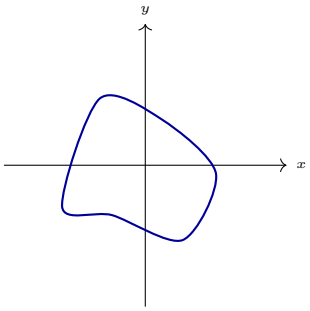
P: _____



Q: _____



R: _____



S: _____

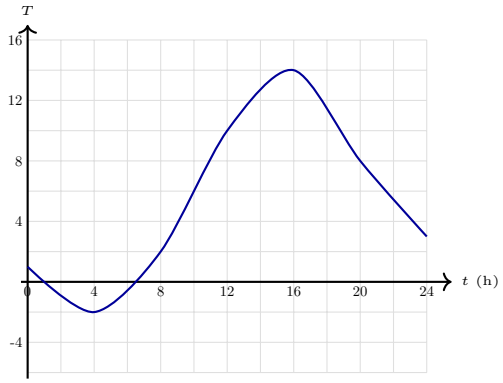
From Graph R, write two ordered pairs that share the same x-value but have different y-values, to show why it is not a function.

5. Functions in Context

Real quantities that are measured over time are functions: at one instant a thermometer, an odometer, or an altimeter can only show one reading. A relation fails when a single input honestly has several outputs (one bus route number, many stops along that route).

Example 5

The graph shows outdoor temperature T (°C) recorded hourly over one day from midnight.



(a) Apply the vertical line test. Is temperature-over-time a function? Explain.

(b) Explain why it makes sense that at one moment in time the outdoor temperature could not be two different values at once.

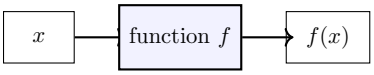
(c) Describe a different everyday relation that would **not** be a function, and explain why in one sentence.

Lesson 9: Function Notation: Evaluating and Interpreting $f(x)$

Renaming y as $f(x)$, then evaluating and interpreting the notation in context.

1. Introducing Function Notation

When a relation is a function (every input has exactly one output), we may rename y with a more descriptive label: $f(x)$, read "f of x." The equations $y = 2x + 5$ and $f(x) = 2x + 5$ say exactly the same thing, but $f(x)$ shows *which* function produced the output and *which* input was used. Warning: $f(x)$ does not mean "f times x" — the brackets hold an input, not a multiplication.



f is a machine: the input x goes in, the rule is applied, and the output $f(x)$ comes out. The letter out front (f, g, h, C, ...) is just the function's **name**; x is the **input** (independent variable) and $f(x)$ is the **output** (dependent variable), the role y always played.

$y =$ notation	function notation
$y = 2x + 5$	$f(x) = 2x + 5$
When $x = 3$: $y = 2(3) + 5$	$f(3) = 2(3) + 5$
$y = 11$	$f(3) = 11$

2. Evaluating a Function for a Given Input

To **evaluate** $f(a)$, replace every x in the formula with a, then simplify using order of operations. Read the direction carefully: $f(3)$ hands you the input 3 and asks for the output, while a statement such as $f(x) = 11$ hands you the output and asks which input produced it (we work backwards like this in Lesson 10).

Example 1

Consider the function $f(x) = 2x + 5$. Evaluate each of the following.

(a) $f(3)$ (b) $f(0)$ (c) $f(-4)$ (d) $f(6)$

Example 2

Consider the functions $f(x) = 5x - 4$ and $g(x) = x^2 + 3$. Evaluate each of the following.
(a) $f(3)$ (b) $g(3)$ (c) $f(-2)$ (d) $g(-1)$ (e) $f(0) + g(0)$

3. Evaluating for a Variable or Expression Input

The same substitution works when the input is not a plain number: to find $f(a)$, $f(k)$, or $f(x + 1)$, replace *every* x with whatever sits inside the brackets and simplify. When the input is an expression, wrap it in brackets everywhere it lands so it is not lost during simplifying.

Example 3

Consider the function $f(x) = 3x + 2$. Determine each of the following.
(a) $f(a)$ (b) $f(k)$ (c) $f(x + 1)$

Example 4

Consider the function $f(x) = x^2 + 4$. Determine each of the following.
(a) $f(a)$ (b) $f(2a)$ (c) $f(x + 3)$
(Hint: first expand $(x + 3)^2$.)

4. Interpreting Function Notation in Context

If $C(n)$ is the total **c**ost, in dollars, of buying n items, then $C(5) = 23$ is shorthand for "buying 5 items costs \$23." The translation runs both ways: "the cost of 8 items is \$35" becomes $C(8) = 35$.

Example 5

A food truck charges a flat \$6 setup fee for a catering order, plus \$3 per taco. The total cost, in dollars, of ordering n tacos is $C(n) = 3n + 6$.
(a) Evaluate $C(4)$. Then explain, in words, what this value represents in this situation.
(b) The statement $C(10) = 36$ is true for this function. Explain, in words, what this statement means in this situation.
(c) Translate into a function notation statement, and check that it is correct: *"The cost of ordering 12 tacos is \$42."*

Lesson 10: Function Notation: Graphs, Conversions, and Real-World Models

Reading $f(x)$ off a graph, switching between $y =$ and $f(x) =$, and modelling real situations.

1. Recall: Evaluating $f(x)$

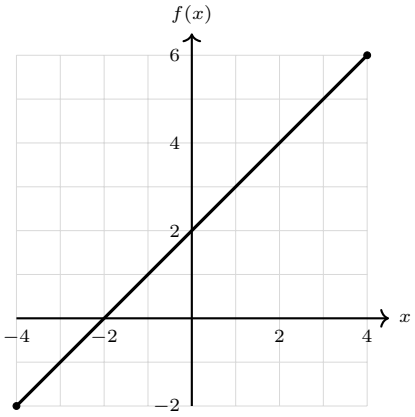
Function notation replaces y with a function name such as $f(x)$, $g(x)$, or $h(t)$. The notation $f(a)$ says: substitute a for the input everywhere it appears in the rule, then simplify. If $f(x) = 3x + 1$, then $f(2) = 3(2) + 1 = 7$.

2. Reading Function Values from a Graph

To find $f(a)$, find a on the horizontal axis, move up or down to the graph, then read across to the vertical axis. To find x when $f(x) = b$, start at b on the vertical axis, move across to the graph, then read up or down to the horizontal axis.

Example 1

The graph of a function f is shown. Use it to answer both parts.



(a) Use the graph to determine **i)** $f(-3)$ **ii)** $f(0)$ **iii)** $f(2)$

(b) Use the graph to determine the value of x if **i)** $f(x) = -2$ **ii)** $f(x) = 3$ **iii)** $f(x) = 6$

3. Converting Between $y =$ Notation and $f(x)$ Notation

The two forms describe exactly the same rule — $f(x)$ and y name the very same output, so $f(x) = 4x + 5$ and $y = 4x + 5$ are the same equation. To convert one way, replace the function name and its bracket, such as $f(x)$, with y; to convert back, replace y with a function name and bracket.

Example 2

(a) Write each in x-y notation. (b) Write each in function notation.

(a) i) $f(x) = 4x - 3$ (a) ii) $g(t) = t + 6$ (b) i) $y = 5x + 2$ (b) ii) $y = 9 - x$

4. Function Notation in Tables

A table can be written in function notation too: the top row lists the inputs x, and the bottom row lists the matching outputs, labelled $f(x)$ or $h(x)$ instead of y. Read a value directly, or fill a missing entry by evaluating the rule.

Example 3

Let $h(x) = 3x + 3$.

(a) Complete the table of values.

x	0	1	2	3	4	5
$h(x)$						

(b) Use your completed table to state the value of $h(4)$.

(c) Use your completed table to determine the value of x for which $h(x) = 15$.

5. Real-World Function Models

Earnings, height, distance, and cost often change by a rule that can be written in function notation. Evaluate the rule to answer the question, then always **interpret** the value in words with the correct units.

Example 4

Jasmine earns money walking dogs. Her earnings are modelled by $E(n) = 8n$, where $E(n)$ is her total earnings, in dollars, for walking n dogs.

(a) Complete the table of values.

n	1	2	3	4	5
$E(n)$					

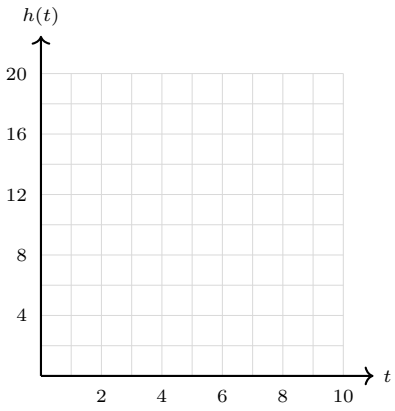
(b) Determine the value of $E(6)$, and explain what it means in this situation.

(c) How many dogs did Jasmine walk if she earned \$56?

(d) Suggest an appropriate domain for this function, given that Jasmine can realistically walk at most 8 dogs in one day.

Example 5

A candle burns down at a steady rate. Its height is modelled by $h(t) = 20 - 2t$, where $h(t)$ is the height, in centimetres, t minutes after it is lit.



(a) Sketch the graph of the function on the grid at the left.

(b) Determine the value of $h(3)$.

(c) Write in words the meaning of $h(3)$.

(d) Evaluate $h(0)$ and explain its significance.

(e) How many minutes will it take for the candle to burn out completely (height $= 0$)?

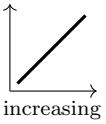
(f) Suggest an appropriate domain and range for the function $h(t)$.

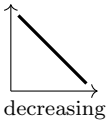
Lesson 11: Interpreting and Sketching Real-World Graphs

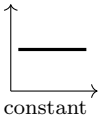
Reading a graph as a story, matching stories to graphs, and sketching your own.

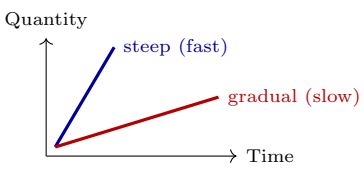
1. Reading the Shape of a Graph

Read a graph left to right in intervals. In each interval the quantity is **increasing** (rising), **decreasing** (falling), or **constant** — *flat*, meaning the quantity is not changing at all (stopped, tap shut off, nothing added or used). Careful: on a **speed** graph, flat means a *steady speed* — only flat at 0 means stopped. The **steepness** of a segment is its **rate of change**: a **steep** segment changes fast, a **gradual** one changes slowly.



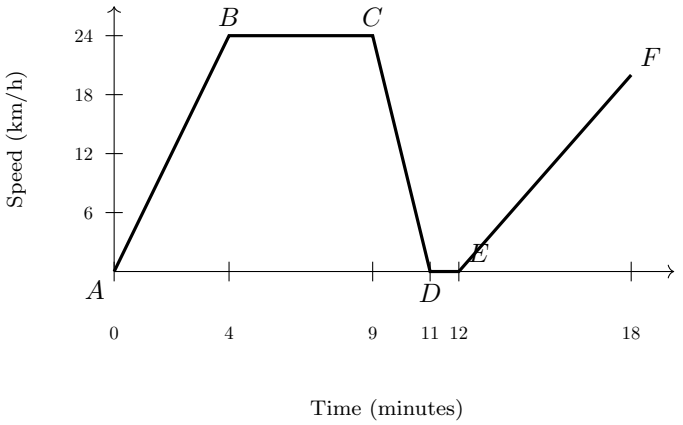






Example 1

Jae rides her bicycle from home to the library along a route with one stop sign. The graph shows her **speed** at every moment of the 18-minute ride, made of five segments meeting at A–F.



(a) With reference to Jae's ride, explain what is happening during each interval.
(i) A to B:

(ii) C to D:

(iii) E to F:

(b) Determine the rate of change, in km/h per minute, between C and D.

(c) Which single segment shows the fastest rate of change (steepest), and what does this tell you about Jae at that time?

(d) For how many minutes, in total, was Jae's speed constant?

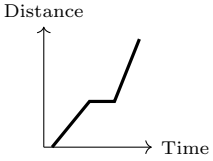
2. Matching a Written Story to a Graph

Break the story into phases, in order, then rule out any candidate graph whose shape contradicts a phase. A stop or a pause is a *flat* segment; "faster" means a *steeper* segment; "coming back" means a *decreasing* one.

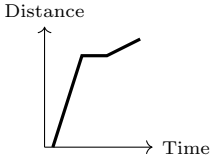
Example 2

Story: Sam leaves home and walks at a steady pace toward school. Partway there, he stops at a corner to talk to a friend for a few minutes. Then he jogs the rest of the way to school, covering the remaining distance *faster* than he walked the first part. Each graph shows Sam's **distance from home** versus **time**.

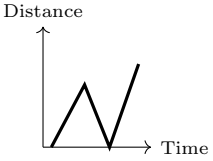
A



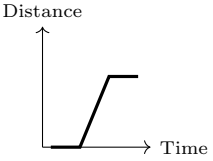
B



C



D



(a) Which graph matches Sam's story?

(b) Explain, in one sentence each, why the *other three* graphs do **not** match.

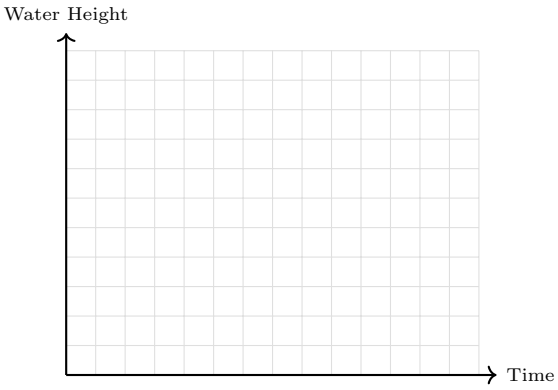
3. Sketching a Graph From a Written Story

With no scale, only the *shape* matters. Choose the axes (time is usually horizontal), split the story into phases in order, and for each phase decide increasing / decreasing / constant and steep / gradual. Join the phases into one continuous graph; exact numbers do not matter.

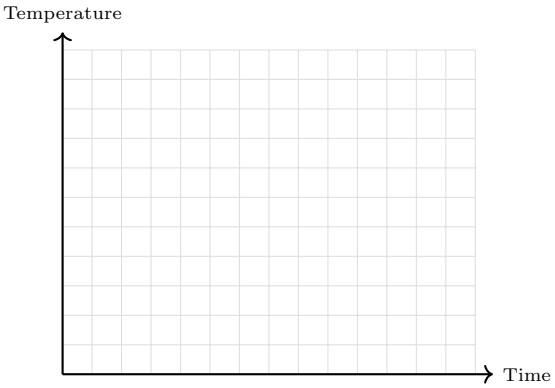
Example 3

Sketch a graph with no scale for each situation. Label your axes.

(a) Height of water in a bottle: a tap fills the bottle at a steady rate from empty until it is full, and then the tap is shut off and the bottle sits full and unchanged.



(b) Temperature of a mug of hot cocoa left on a table: it cools quickly at first, then cools more and more slowly as it gets close to room temperature (never quite reaching it).

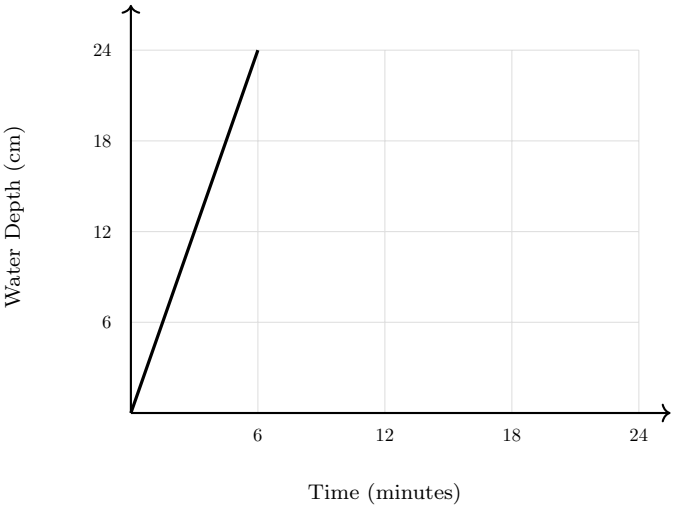


4. Combining Reading and Sketching

The richest questions graph *part* of a story, ask you to interpret that part (often its rate of change), and then have you continue the sketch for the rest of the story on the same grid.

Example 4

Mia runs a bath. For the first 6 minutes the tap fills the tub at a steady rate and the depth rises from 0 cm to 24 cm — already drawn below. She then soaks for 10 minutes, during which the depth stays about the same. Finally she pulls the plug and the tub drains at a steady rate, from 24 cm back to 0 cm over the next 8 minutes.



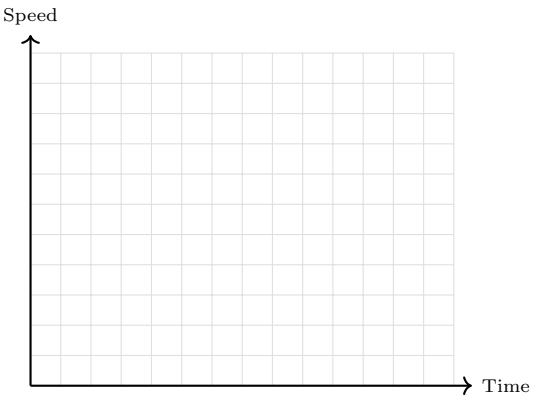
(a) What is the rate of change of the water depth during the first 6 minutes (while the tap is filling the tub)?

(b) On the same grid, continue the sketch to show the rest of Mia's bath (the soak, then the drain).

Example 5

A family drives from home to a mountain cabin. Sketch their **speed** versus **time** for the whole trip.

- They pull out of the driveway and speed up steadily to highway speed.
- They cruise at that highway speed for a long stretch.
- They slow down and come to a complete stop at a gas station, where they stay stopped for a while to refuel.
- After refuelling, they speed up again, but only back up to a *slightly slower* cruising speed than before (more traffic near the cabin).
- Finally, they slow down and come to a stop as they arrive at the cabin.



Lesson 12: Unit Review: Relations and Functions

One reminder and one review question for every idea in Unit 5, then mixed challenges.

Quick Reference — the Whole Unit on One Card

Cartesian plane — (x, y): right/left, then up/down. I$(+,+)$, II$(-,+)$, III$(-,-)$, IV$(+,-)$; a zero coordinate puts the point on an axis.

Intercepts — x-intercept: $y = 0$. y-intercept: $x = 0$.

Function — every input has exactly one output; on a graph, no vertical line may cross twice.

Relation — one set of pairs shown four ways: ordered pairs, table of values, mapping diagram, graph.

Domain / Range — domain = all x-values; range = all y-values. Use { } for lists, inequalities for segments.

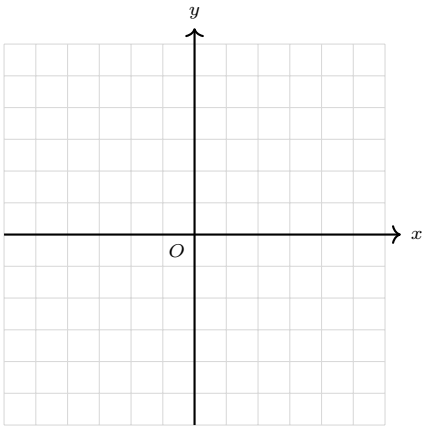
Function notation — $f(a)$: substitute a for x. Given $f(x) = b$: set the rule equal to b and solve.

1. The Cartesian Plane and Plotting Points

Signs of the coordinates decide the quadrant; a zero coordinate puts a point on an axis.

Example 1

Answer part (a) below; plot and label part (b) on the grid.



(a) State the quadrant (or axis) containing each point.

$P(-6, 2)$ _____ $Q(4, -5)$ _____

$R(0, 7)$ _____ $S(-3, -8)$ _____

(b) Plot and label $T(3, 5)$, $U(-4, 1)$, $V(-2, -6)$ and $W(5, -3)$ on the grid.

2. Patterns and Variables

A growing pattern adds the same amount each step; the variable you control is independent, the one that results is dependent.

Example 2

A community centre sets up rows of chairs. Row 1 has 6 chairs, and each row after has 5 more than the row before.

(a) Complete the table of values for Rows 1 to 5.

Row Number, n	1	2	3	4	5
Chairs, C	6				

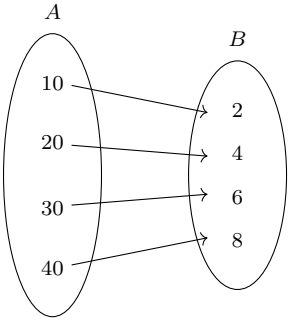
(b) State the independent variable and the dependent variable.

3. Representing a Relation

Ordered pairs, a table, a mapping diagram and a graph are four pictures of the same set of pairs.

Example 3

The mapping diagram shows a relation between sets A and B.



(a) Write this relation as a set of ordered pairs (A, B).

(b) Represent the same relation in a table of values.

4. Intercepts

An x-intercept has $y = 0$; a y-intercept has $x = 0$.

Example 4

The table shows several points on a relation.

x	-1	0	1	2	3	4
y	-4	-3	-2	-1	0	1

(a) Which ordered pair is the y-intercept?

(b) Which ordered pair is the x-intercept?

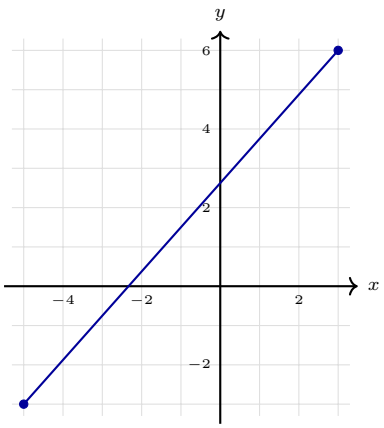
5. Domain and Range

Domain = all the x-values; range = all the y-values — read off a graph, or listed in set notation.

Example 5

State domain and range for each relation.

(a) The graph shows a line segment. State its domain and range.



(b) State the domain and range, using set notation, of the relation $(3, 7),\ (5, 2),\ (3, 9),\ (8, 2)$.

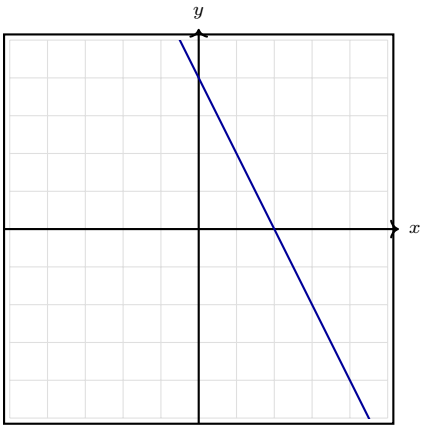
6. Using Technology to Graph a Relation

A graphing calculator draws the relation once you enter it and set a window; a line running off both edges without turning has domain and range of all real numbers.

Example 6

Diego graphs $y = -2x + 8$ in the standard window, $x : [-10, 10]$, $y : [-10, 10]$.

(a) State the window shown on the screen.



(b) Use the screen to estimate the x-intercept and the y-intercept.

(c) State the domain and range of this relation.

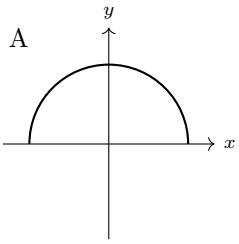
7. Functions and the Vertical Line Test

A relation is a function when each input has exactly one output — no vertical line may cross the graph twice.

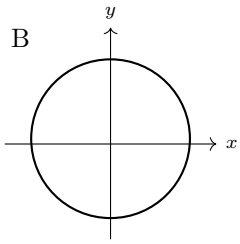
Example 7

Exactly one graph below is **not** a function. Use the vertical line test to identify it, and justify your answer.

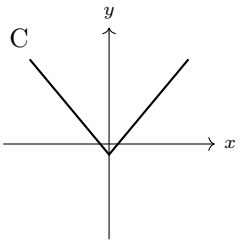
A



B



C



8. Evaluating Function Notation

To find $f(a)$, substitute a for every x; to work backward, set the rule equal to the output and solve.

Example 8

Consider the function $g(x) = 4x - 9$.

(a) Determine the value of $g(6)$.

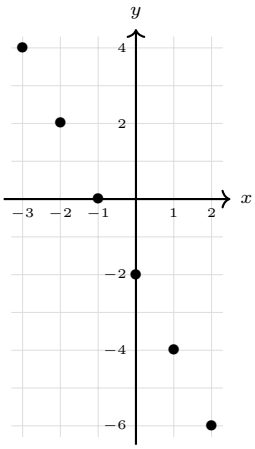
(b) If $g(k) = 7$, determine the value of k.

9. Function Notation with Graphs and Models

Function notation can be read off a graph, or used to model a real cost that depends on a quantity n.

Example 9

Use the graph for (a); use the model for (b).



(a) The diagram shows the graph of $y = f(x)$. Use it to determine the value of $f(-1) + f(2)$.

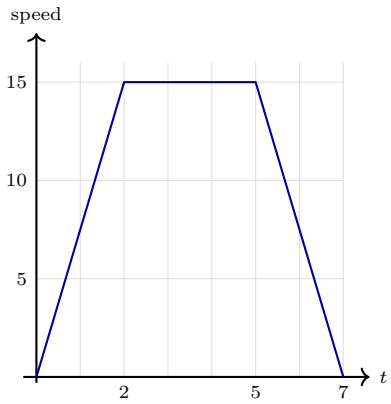
(b) A dog-walking business charges a \$10 booking fee plus \$8 for each walk, so the cost of n walks is $W(n) = 10 + 8n$. Determine $W(5)$, and state the domain of this relation.

10. Interpreting and Sketching Real-World Graphs

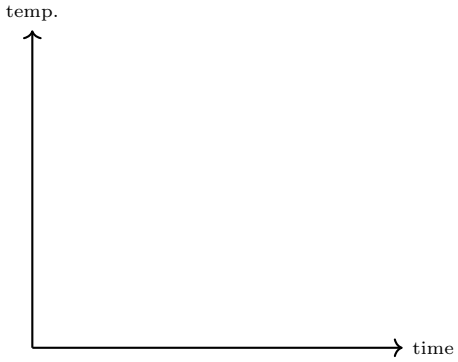
Rising means increasing, falling means decreasing, flat means staying the same — no numbers required.

Example 10

Read one graph; sketch the other.



(a) The graph shows a cyclist's speed, in km/h, during the first 7 minutes of a ride. Describe, in words, what happens to the speed during the first 2 minutes, during minutes 2 through 5, and during the last 2 minutes.



(b) On the blank grid, sketch a graph showing the temperature of a cup of hot coffee, from the moment it is poured until it has cooled down to room temperature and stays there.

11. Mixed Review: Putting It All Together

Each question below combines more than one skill — exactly like a test question.

Challenge 1

A landscaping company charges a \$25 site fee plus \$15 per tree planted: $C(p) = 25 + 15p$.

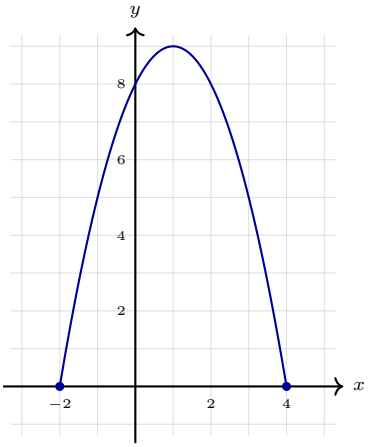
(a) Determine $C(8)$.

(b) If $C(p) = 175$, determine the number of trees planted.

(c) State an appropriate domain for this function, given the context.

Challenge 2

The graph below shows a relation.



(a) Is this relation a function? Justify your answer using the vertical line test.

(b) State the domain and range of the relation.

(c) State the coordinates of the x-intercepts and the y-intercept.

Challenge 3

A community garden has 8 plants in Week 1, and 6 more plants are added every week after that. Let n be the week number and $P(n)$ the number of plants.

(a) Write a function rule $P(n)$.

(b) Determine $P(6)$.

(c) In what week will there be 62 plants?

(d) State a reasonable domain, given the context.